STAT/MA 41600
In-Class Problem Set #33: November 4, 2026
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Problem Set 33 Answers

1a. The random variable Y is a Gamma random variable.
1b. The parameters of Y are $r = 10$ and $\lambda = 1/3$.
1c. We have $E(Y) = r/\lambda = (10)(3) = 30$.
1d. We have $\text{Var}(Y) = r/\lambda^2 = (10)(3^2) = 90$.

2. No, Y is not a Gamma random variable. Although the exponential random variables X_1, X_2, X_3 are independent, their parameters (i.e., their respective values of λ) are different from each other.

3a. The random variable Y is a Gamma random variable with $E(Y) = (3)(1/5) = 3/5$ and with $\text{Var}(Y) = (3)(1/5)^2 = 3/25$.
3b. The pdf of Y is $f_Y(y) = \frac{5^3}{(3-1)!} y^{3-1} e^{-5y}$ for $y > 0$, and $f_Y(y) = 0$ otherwise.

4a. We have $\text{Var}(24Y) = 24^2 \text{Var}(Y) = (24^2)(90) = 51840$.
4b. We have $\text{Var}(Y/8) = (1/8)^2 \text{Var}(Y) = (1/8)^2(3/25) = 3/1600$.
4c. Because the X_is are independent, we have $\text{Var}(6X_1 + 7X_2 + 8X_3) = 6^2 \text{Var}(X_1) + 7^2 \text{Var}(X_2) + 8^2 \text{Var}(X_3) = (6^2)(1/5)^2 + (7^2)(1/5)^2 + (8^2)(1/5)^2 = 149/25$.
4d. No, W is not a Gamma random variable. Although the three random variables $6X_1$ and $7X_2$ and $8X_3$ are exponential random variables, their parameters (i.e., their respective values of λ) are different from each other, i.e., 5/6 and 5/7 and 5/8.

5. There are 4 available types of witches' brew (bubbling, smoky, acidic, and slimy). Upon drinking the potions, each person will begin to feel the effects of the potion in an exponential time with mean 2 minutes. Suppose that 4 people each simultaneously drink one of the witches' brews, and their times to feel the effects of the potions are independent random variables X_1, X_2, X_3, X_4.

The random variable Y is a Gamma random variable with parameters $r = 4$ and $\lambda = 1/2$. It follows that the pdf of Y is $f_Y(y) = \frac{(1/2)^4}{(4-1)!} y^{4-1} e^{-y/2}$ for $y > 0$, and $f_Y(y) = 0$ otherwise.

Mind bender (not required). All 31! arrangements of the random variables are equally likely. There are 31 possible rankings of X_1 compared to the other random variables. There are 16 possible locations (among the 31 random variables) where X_1 can be placed, so that X_1 is bigger than an even number of the 30 other random variables. Please note that the parameter of the exponential random variables does not actually matter at all! It only matters that the probability of any ties is 0, and that all possible orderings are equally likely (because the variables are independent and identically distributed).